Hot Weather Alert Windows

Coding Assignment - Problem 4 | Python

```
Task: Count length-k contiguous windows whose average is at least threshold.
Compare window_sum >= k*threshold to avoid decimal division.
def countAlerts(readings, k, threshold):
window_sum = sum(readings[:k])
count = int(window_sum >= k * threshold)
for right in range(k, len(readings)):
window_sum += readings[right] - readings[right-k]
if window_sum >= k * threshold:
count += 1
return count

print(countAlerts([2,2,2,2,5,5,5,8], 3, 4))

Output: 3
Recalculating each window costs O((n-k+1)*k), worst-case O(n*k).
Sliding window: O(n) time and O(1) auxiliary space.
```